Dexpirona Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-986/B

Reference Standards

Primary reference standards for the drug substance and specified impurities are qualified against compendial standards where available. Working standards are qualified against the primary standard under an approved protocol and are stored under controlled conditions.

Deviations observed during the campaign were investigated and closed with no impact to product quality. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Statistical evaluation of batch release data demonstrates process capability well within specification limits. The control strategy links critical quality attributes to critical process parameters identified during development. Cleaning verification swab results were below the health-based exposure limits derived for the product.

No changes to the approved analytical procedures are proposed in this submission. Deviations observed during the campaign were investigated and closed with no impact to product quality. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement.

Regional Information

No excipients of human or animal origin are used in the drug product with the exception of lactose monohydrate, for which a TSE/BSE compliance declaration is provided in Module 3.2.A.2. Certificates of suitability are maintained where applicable.

Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Cleaning verification swab results were below the health-based exposure limits derived for the product.

Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Deviations observed during the campaign were investigated and closed with no impact to product quality.

Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

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Control of Excipients (3.2.P.4)

For the commercial formulation F-986/B, Plasdone K-29/32 (Povidone, binder) is used.

Selection Justification

The selection of Plasdone K-29/32 is justified by compatibility with the active ingredient demonstrated in binary stress studies observed during development studies.

Grade selection and control follow the principles of ICH Q6A.

Commercial Control Strategy — Excipients

Component	Function	Grade per current specification
Povidone	binder	Plasdone K-29/32
Reference		SPEC-EX-4774 Rev 04

Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

No changes to the approved analytical procedures are proposed in this submission. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results.

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Manufacturing Process Overview

The drug product is manufactured by a conventional wet granulation process comprising dispensing, high-shear granulation, fluid-bed drying, milling, blending, compression, and film coating. In-process controls are applied at each unit operation, including loss on drying after fluid-bed drying and blend uniformity prior to compression. Process parameters were established during process performance qualification and are maintained under the site quality system.

No changes to the approved analytical procedures are proposed in this submission. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions.

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Analytical Method Validation Summary

The assay and related-substances procedures employ reversed-phase high performance liquid chromatography with ultraviolet detection. Validation demonstrated acceptable specificity, linearity, accuracy, precision, and robustness. Forced degradation studies confirmed the stability-indicating capability of the related-substances method.

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Facilities and Equipment

Manufacturing is performed at the approved commercial site using dedicated product-contact equipment trains. Equipment qualification and cleaning validation are maintained under the site validation master plan. No changes to the approved facility registration are proposed.

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